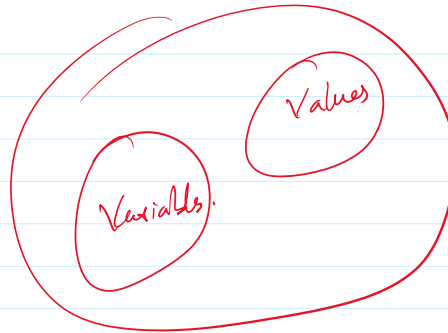
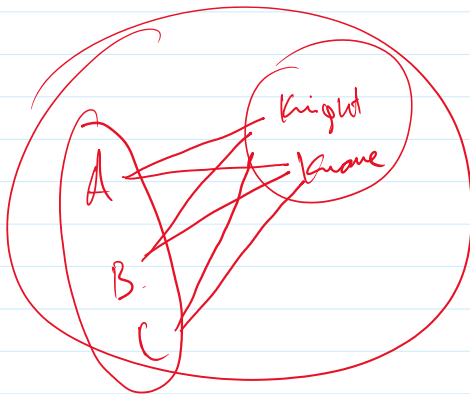


lecture 4:-

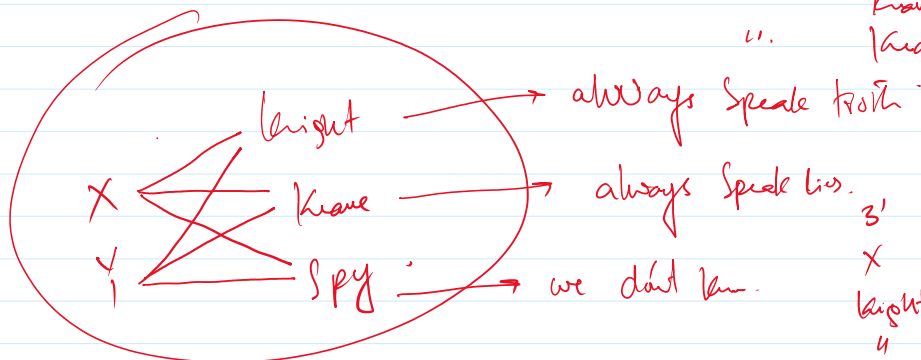
$$2^3 = 8.$$

Values Variables.



$p \equiv A \text{ is a Knight} \rightarrow \neg p \equiv A \text{ is a Knaves}$
 $q \equiv B \text{ is a Knight} \rightarrow \neg q \equiv B \text{ is a Knaves}$
 $r \equiv C \text{ is a Knight} \rightarrow \neg r \equiv C \text{ is a Knaves}$

2^2	2^1	2^0
A	B	C
Knight	Knight	Knight
"	Knight	Knaves
"	Knaves	Knight
"	Knaves	Knaves
Knaves	Knight	Knight
"	Knight	Knaves
"	Knaves	Knight
"	Knaves	Knaves



always speaks truth.

always speaks lies.

we don't know.

Let $p_1 \equiv X \text{ is a Knight} \rightarrow \neg p_1 \equiv X \text{ is not a Knight}$
 $p_2 \equiv X \text{ is a Knaves} \rightarrow \neg p_2 \equiv X \text{ is not a Knaves}$
 $p_3 \equiv X \text{ is a Spy} \rightarrow \neg p_3 \equiv X \text{ is not a Spy}$

$q_1 \equiv Y \text{ is a Knight}$
 $q_2 \equiv \text{" " " Knaves}$
 $q_3 \equiv \text{" " " Spy}$

2^1	2^0
X	Y
Knight	Knight
"	Knaves
"	Spy
Knaves	Knight
"	Knaves
"	Spy
Spy	Knight
"	Knaves
"	Spy

?

$\neg q_1 \vee q_2$ " " " " know -
 $q_3 \vee \neg q_1$ " " " " Spy
 ? " know
 " Spy.

X Says "I am a Spy or Y is a knight!" ✓ $P_3 \vee q_1$

Y Says " " ✓

X Knight
 Y Knight
 $\neg T$
 $\neg T$

$P_1 \vee T$
 $P_2 \vee F$
 $P_3 \vee F$
 $\neg P_1 \vee F$
 $\neg P_2 \vee T$
 $\neg P_3 \vee T$

$q_1 \vee T$
 $q_2 \vee F$
 $q_3 \vee F$
 $\neg q_1 \vee F$
 $\neg q_2 \vee T$
 $\neg q_3 \vee T$

X Knight
 Y Spy
 $\neg T$
 $\neg ?$

Propositional logic

Assignment #2

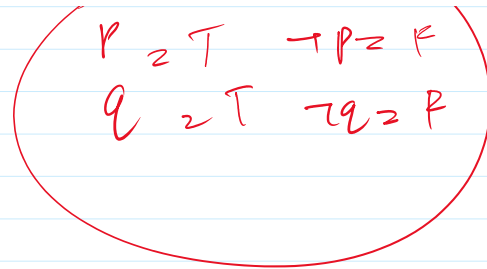
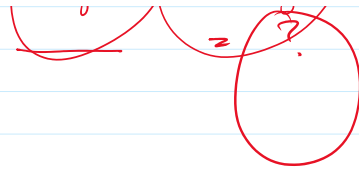
Do Ex 55-59
Page 20.

8-Sep-2025.

Handwritten. → Box — Do it back

A Knight
 B Knight
 $\neg ?$

$P \vee T$
 $\neg P \vee F$
 $\neg T$

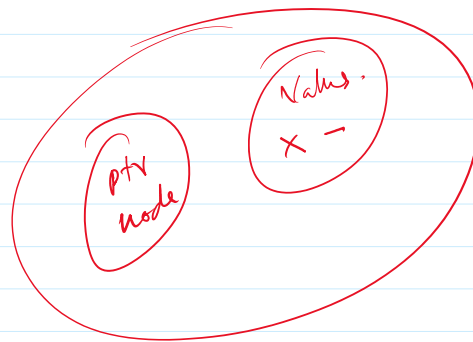


Handwritten text: $\frac{P}{Q} \mid \frac{P}{Q}$

Handwritten text: $\frac{X}{P} \mid \frac{X}{Q}$

A Says
B Say nothing

Handwritten text: $\frac{X}{P} \mid \frac{X}{Q}$



+ 2 ~~truth~~ lies.
- 2 lies.

Quiz #2

05-Sep-2025 -

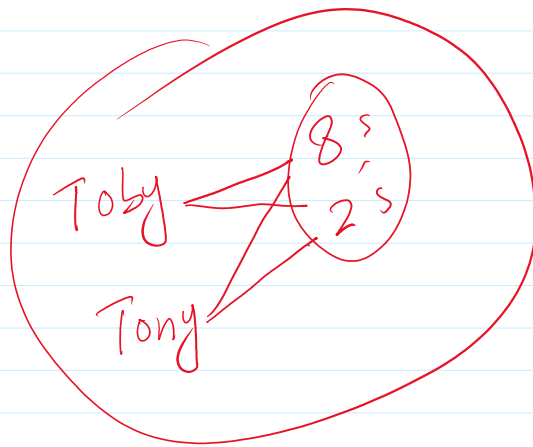
Two brothers, **Toby** and **Tony**, are either 8s or 2s. 8s only speak truth when it is Friday. 2s always speak lies

Toby says: "I am 8s whenever Tony is 2s"

Tony says nothing.

Determine Toby and Tony types

Solution -



Let $p \equiv$ Toby is 8's $\neg p \equiv$ Toby is 2's
 $q \equiv$ Tony is 8's $\neg q \equiv$ Tony is 2's

Toby is 8s whenever Tony is 2s
 p $\neg q$

$$\neg q \rightarrow p$$

Toby Tony
 8 8
 $\neg q \rightarrow p = T$
 $p \rightarrow T = T$
 $T \rightarrow T = T$ ✓

$p = T$ $\neg p = F$
 $q = T$ $\neg q = F$

Toby Tony
 8 2
 $\neg q \rightarrow p = T$
 $T \rightarrow T = T$ ✓

$p = T$ $\neg p = F$
 $q = F$ $\neg q = T$

Toby Tony
 2 8
 $\neg q \rightarrow p = F$
 $p \rightarrow F = F$

$p = F$ $\neg p = T$
 $q = T$ $\neg q = F$

$T \neq P$. Does not hold.

Toby Tony

20 20

$P \neq P$

$\neg P \neq T$

$\neg q \rightarrow P \neq P$.

$q \neq P$

$\neg q \neq T$.

$T \rightarrow P \neq P$

$P \neq P$ Case hold.

we are unable to fill the types.